As Predicted:"*(July 2021) Self-Punishing for Fortuitous Reward – BRET*" (#69978)

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1) Have any data been collected for this study already?
No, no data have been collected for this study yet.

2) What's the main question being asked or hypothesis being tested in this study?
We theorized that choosing to self-inflict (vs. to avoid) suffering would lead to greater risk-taking decisions.

More specifically, we predict that the CHANGE in risk-taking decision (from before to after making the decision to self-inflict/avoid suffering) would be MORE POSITIVE among participants who choose to self-inflict suffering compared to those who choose to avoid suffering.

3) Describe the key dependent variable(s) specifying how they will be measured.
The key dependent variable is participants' decision on two rounds of the Bomb Risk Elicitation Task (BRET). In each round of this task, participants are presented with 25 boxes, one of which contains a bomb. Participants can choose to open any number of boxes (ranging from 0 to 25). If none of the boxes they open contains the bomb, they will earn a fixed bonus payment for each box they open. If one of opened boxes contains the bomb, they will not receive any bonus payment. In short, the key dependent variable is the number of boxes they choose to open in the BRET.

Participants in the 'Choice condition' (see next section for more information about condition) will additionally make a choice between two versions of a task that purportedly measures 'finger agility' (hereafter 'version choice'). One of the two versions will ostensibly cause mild and temporary hand pain ('Pain' group) while the other does not involve any mention of pain ('No pain' group). Choosing the painful version is indicative of the tendency to self-inflict suffering. This measure is adapted from a prior study by Slepian and Bastian (2017). 'Version choice' serves as a non-manipulated between-subject grouping variable. Participants in the 'No choice' condition do not make such a choice.

The order in which the dependent variables will be measured is as follows. Participants will first make decision on the first round of the BRET (BRET #1). Next, participants in the 'Choice condition' will indicate their 'version choice' while participants in the "No choice' condition will read about the finger agility task but not make any choice. Next, all participants will make decision on the second round of the BRET (BRET #2).

4) How many and which conditions will participants be assigned to?
Participants will be randomly assigned to the 'Choice' and 'No choice' between-subject conditions (with an allocation ratio of 2:1).

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.
Our primary analysis would be a 2 (Time: BRET #1 vs. BRET #2) x 2 (Version Choice: Pain vs. No Pain) mixed ANOVA with decisions on the BRET task (i.e., number of boxes opened) as the dependent variable. We predict a significant interaction effect such that the change from BRET #1 to BRET #2 would be more positive in the 'pain' group compared to the 'no pain' group.

We will also conduct additional exploratory analyses involving (i) the 'No choice' condition, and (ii) participants' belief in karma.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.
Towards the end of the study, participants in the 'Choice condition' will be asked to recall the version of the task they had previously chosen. Only participants who provide a correct response will be included in confirmatory analyses.

7) How many observations will be collected or what will determine sample size?
No need to justify decision, but be precise about exactly how the number will be determined.
Our target sample size is 500. Because the study will be conducted on MTurk, the actual sample size may deviate very slightly from our target due to reasons outside of our control.

Sensitivity power analysis conducted using G*Power indicates that a sample size of 450 (150 per cell), with a correlation between repeated measure at r = .05, will confer 80% power to detect a the between-within interaction effect (for the mixed ANOVA outlined in the analysis section) of f = .08 at an alpha level of .05. We deem this level of power to be adequate. We slightly increased our target sample size by 50 (to 500) to account for a small portion of participants who may not meet our inclusion criteria.

8) Anything else you would like to pre-register?
(e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)
For exploratory purposes, we also ask the following questions:
o Self-reported score on the finger agility task
o The level of pain participants actually experienced in the task
o Demographic information (age, gender)
o 16-item Belief in Karma scale

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